$$\frac{(\alpha+\beta)|00\rangle + (\alpha-\beta)|11\rangle}{\sqrt{2}}$$

before measurement

lift, then V

record 1 (safe, $g = |0\rangle$)

$$\frac{\alpha+\beta}{\sqrt{2}} \cdot |0\rangle \otimes |0\rangle$$

weight $|\alpha+\beta|^2/2$

record 0 (unsafe)

$$\frac{\alpha-\beta}{\sqrt{2}} \cdot \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

weight $|\alpha-\beta|^2/2$